

CS & IT ENGINEERING



Digital Logic

Sequential Circuits

Lecture No 01



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Recap of Previous Lecture



Topic



Combinational Circuits

Topics to be Covered



Topic

Sequential Circuits



* Combinational Circuits $\rightarrow$ (V.V. Imp)

$\hookrightarrow$ 1) 5 Step framework:

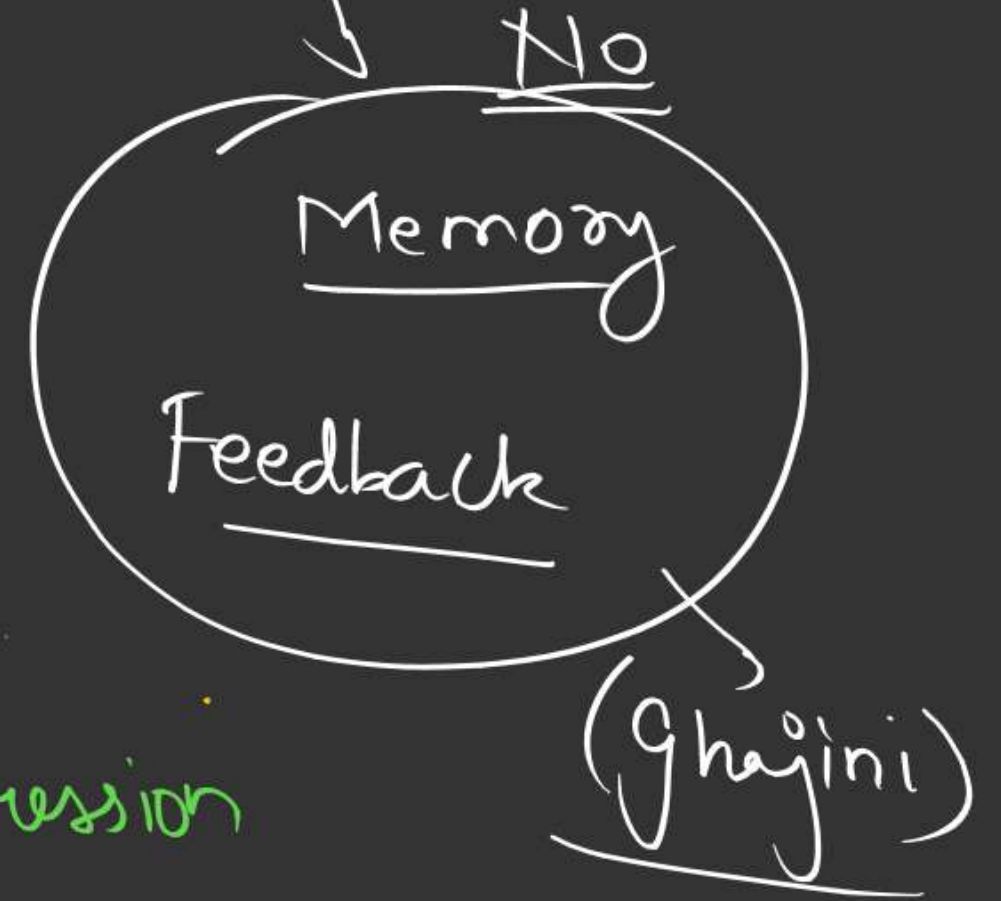
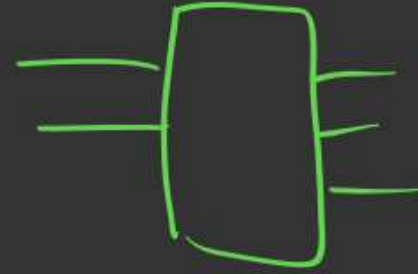
$\hookrightarrow$ Step 1:- no. of i/p & o/p.

Step 2:- Truth Table

Step 3:- Logical Expression

Step 4:- Minimize logical Expression

Step 5:- Circuit Design / Hardware Implementation



(V.V. Imp)

- 1) logical properties
- 2) K-map.

1) Comparator (magnitude)

- 1 bit
- 2 bit, 3 bit
- generalised (semi-minimized exp^s)

(V.V. Imp)

2) MUX

- Type 1: Larger MUX using smaller MUX
- Type 2: MUX as a Universal Circuit (2x1 MUX)
- Type 3: Minimization. 
- Type 4: Cascading multiple MUX
- Type 5: Function $\rightarrow$ MUX design.
- Type 6: Delay in MUX.

3) Demux
↳ designing

4) Encoder (any code $\rightarrow$ Binary)
↳ Priority Encoder.

5) Decoder (Binary $\rightarrow$ any code)

6) Adder & Subtractor

Adder

↳ sum
↳ carry

HA
FA
HS
FS

Subtractor

↳ Difference
↳ Borrow.

NAND
NOR

FA $\rightarrow$ 2 HA + OR

FS $\rightarrow$ 2 FS + OR

FA/FS $\rightarrow$ 9 NAND/NOR

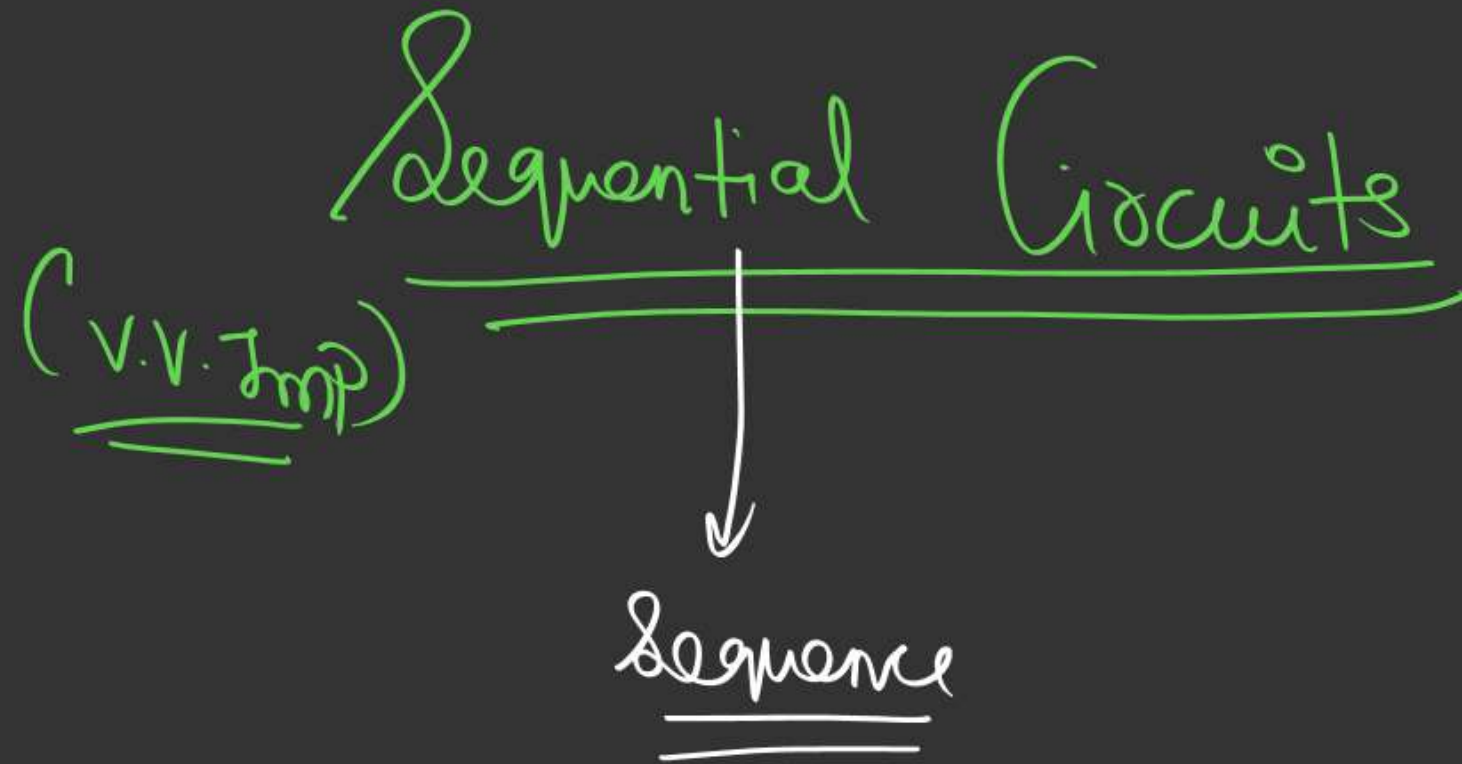
HA/HS $\rightarrow$ 5 NAND/NOR

Problem Solving

7/ Serial Adder

↳ Remaining addrs in Tomorrow lecture

(9 AM) only

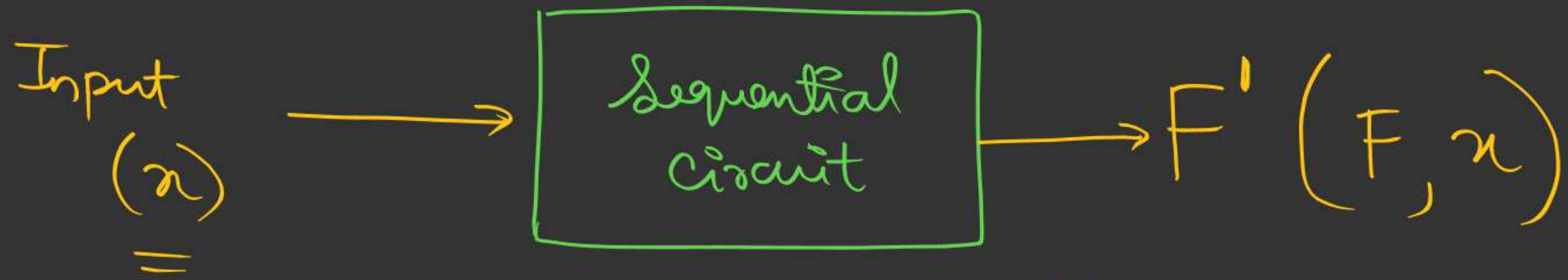


- ✓ 1) Latches
- ⑤ 2) Flip-Flops
- 3) Registers
- 4) Counters

Sequential Circuit:

1) A circuit that has both feedback as well as memory.

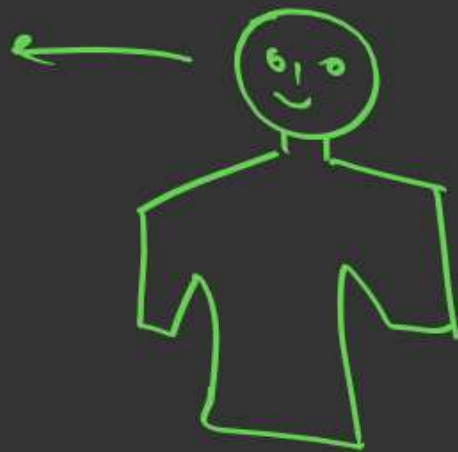
② ★ Current output of the Sequential Circuit depends on both: the current/present state of the i/p as well as the previous Output.



$F \rightarrow$ previous output
 $x \rightarrow$ input.

eg1

You

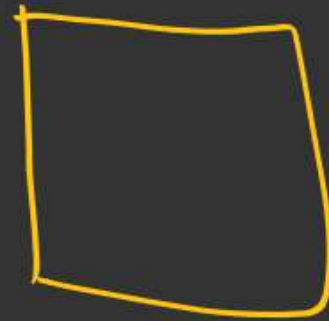
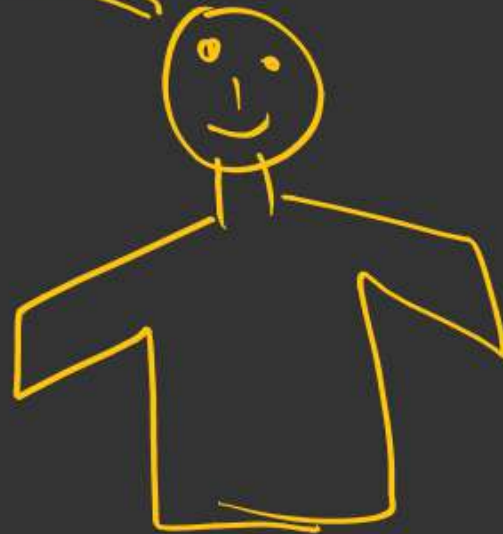
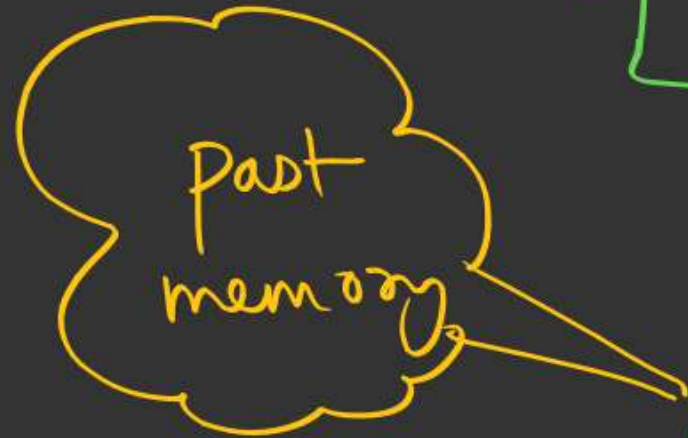


Engineering
Assignment



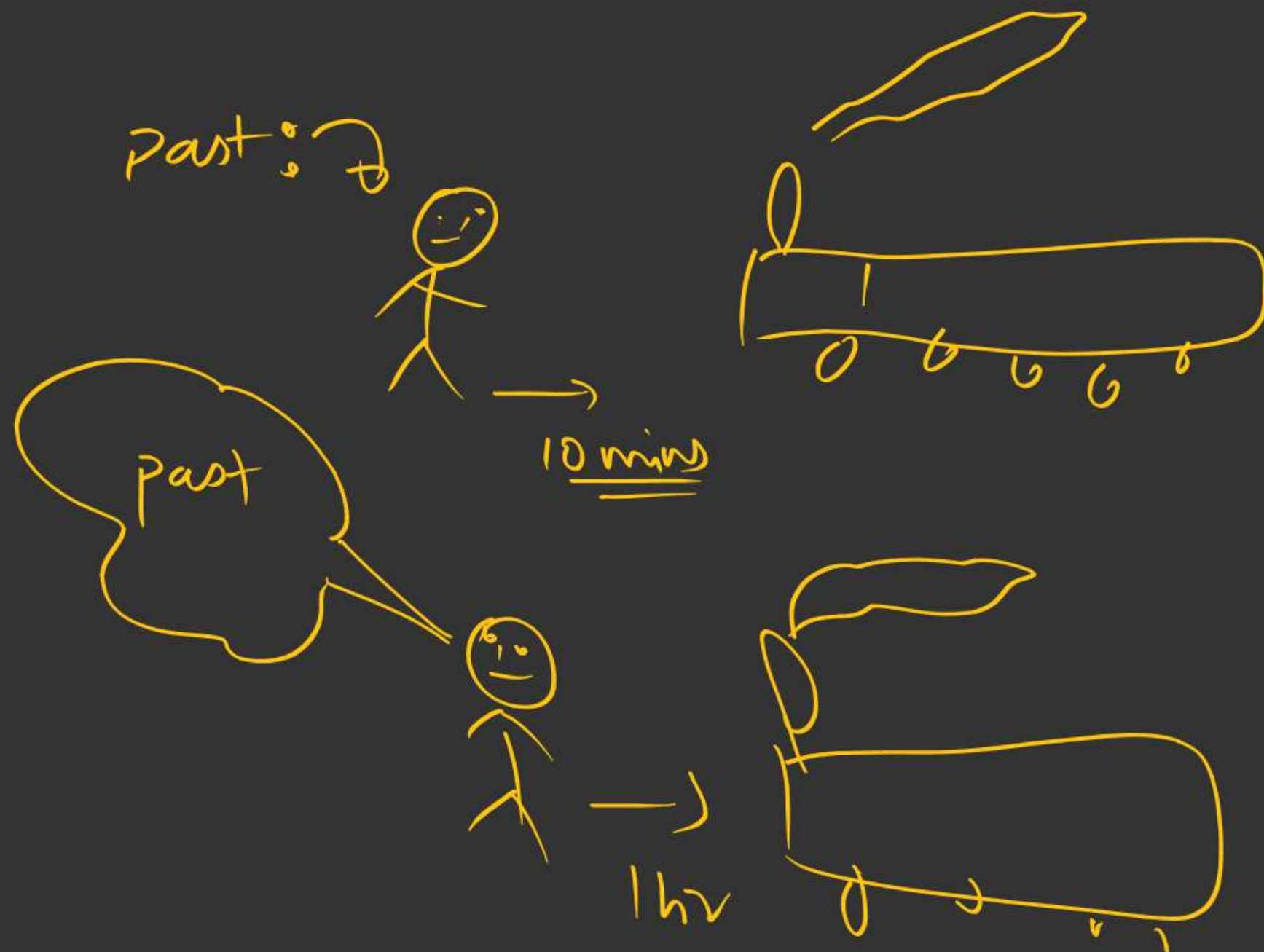
1 week left

unable to
complete
at last moment

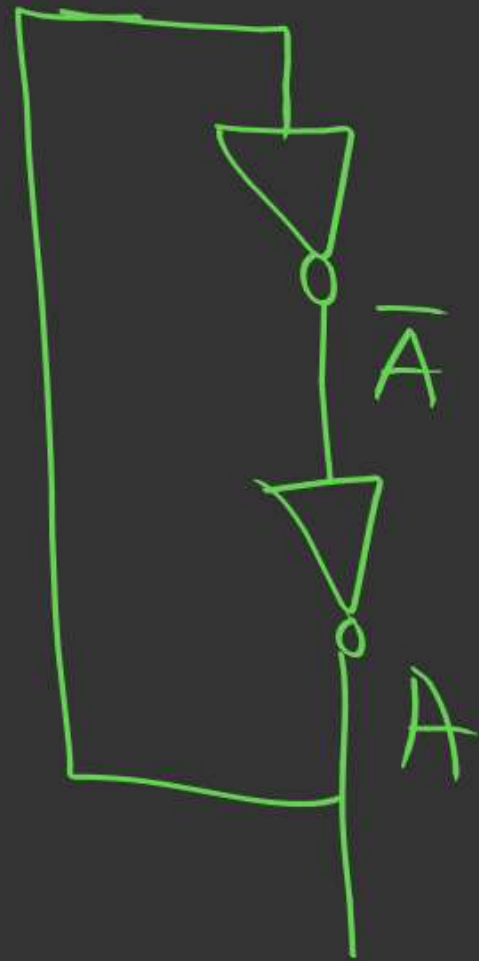


1 week left

eg2

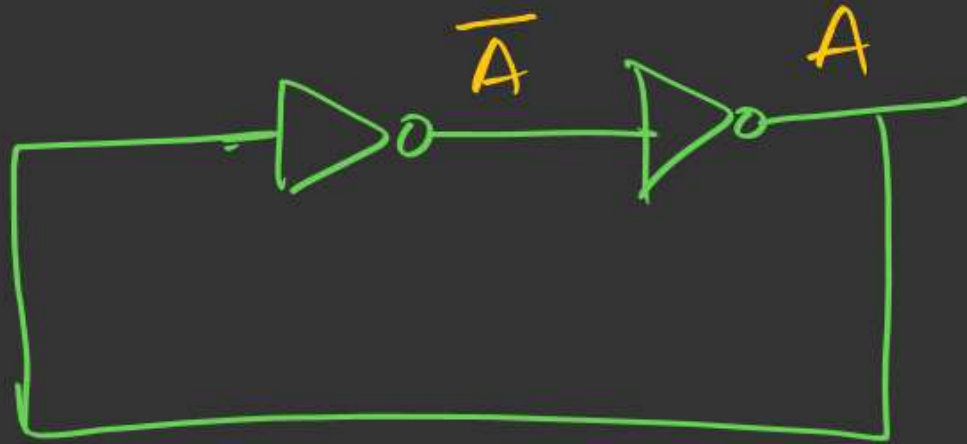


Basic memory element:

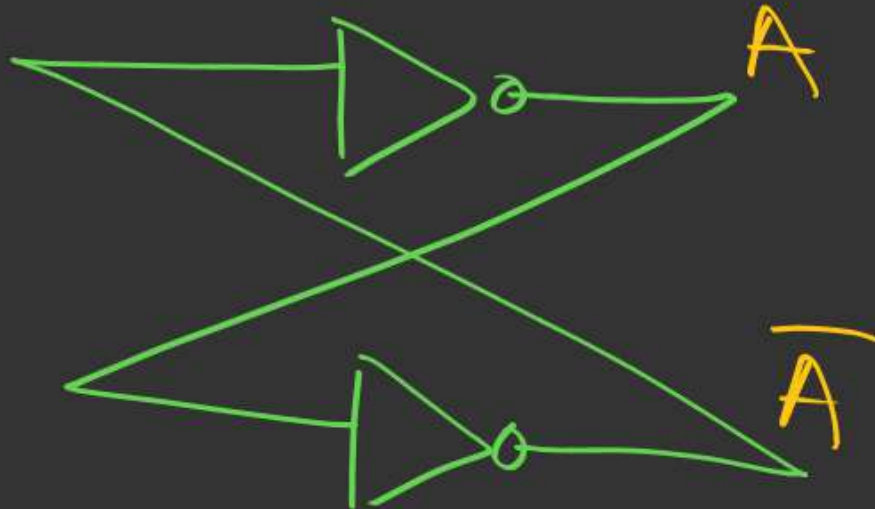


(Q)

A)



B)



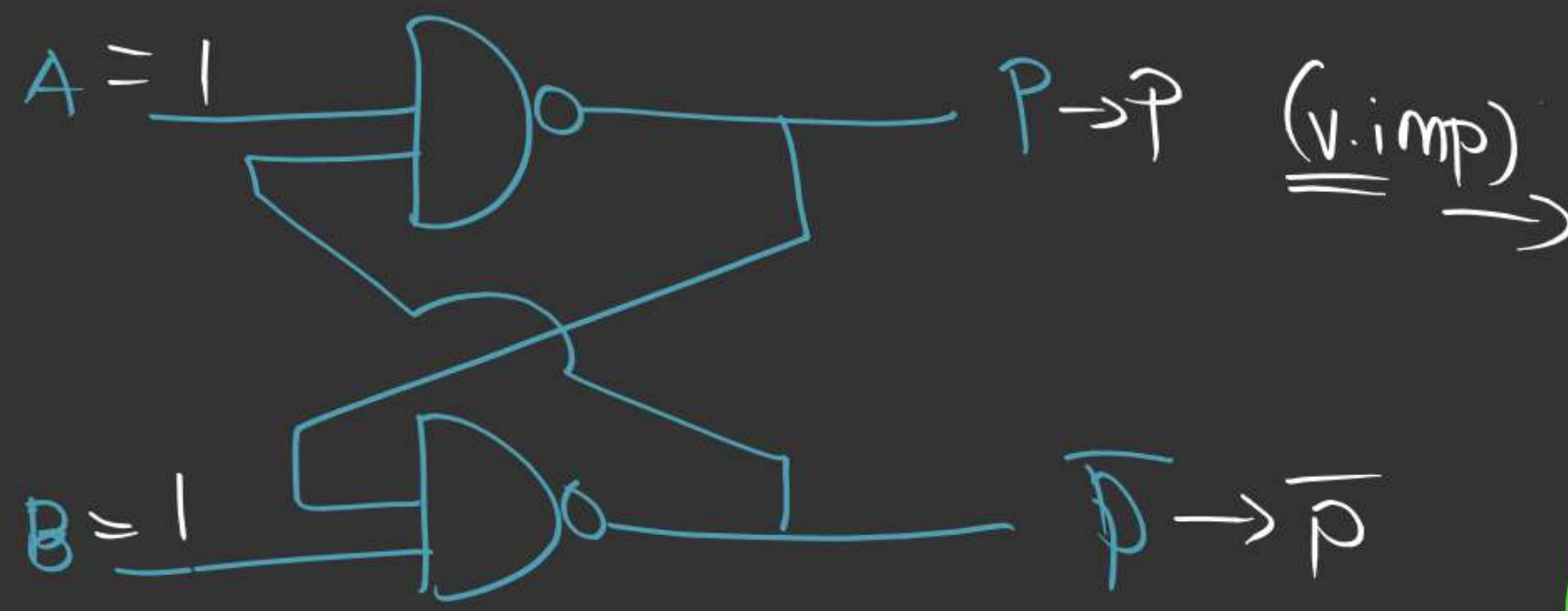
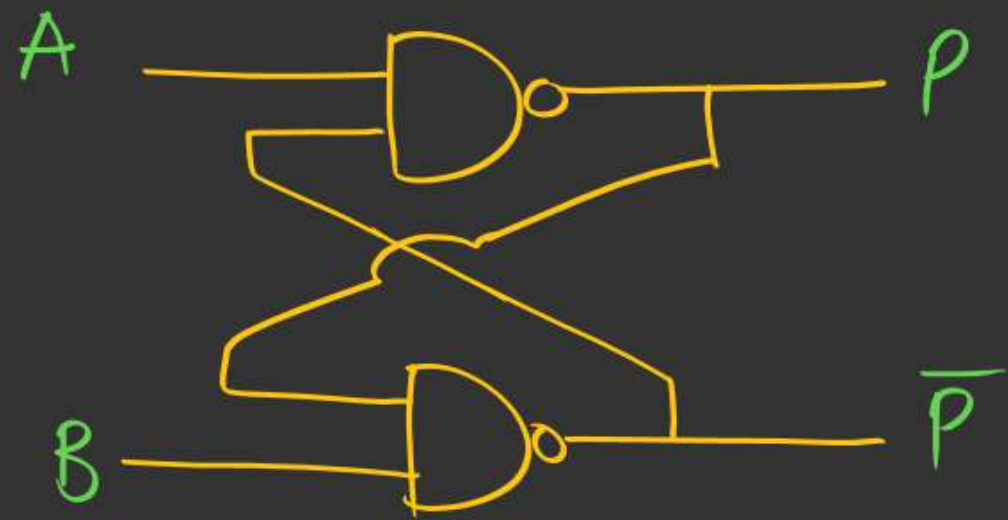
1) Latches

(Basic memory element)

Logically
Same or diff?



Same ✓



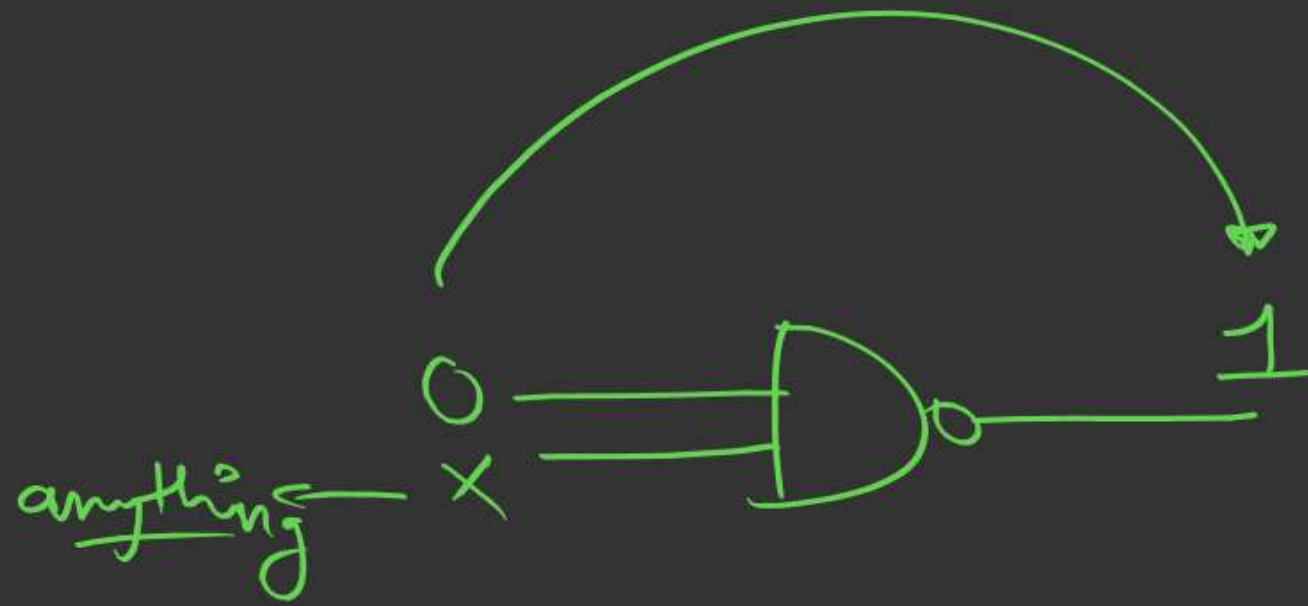
<u>A</u>	<u>B</u>	<u>P</u>	<u>$\bar{P}$</u>
0	0	1	1
0	1	1	0
1	0	0	1
1	1	P	$\bar{P}$

invalid

HOLD

previous
o/p holds.
curr o/p =
prev o/p

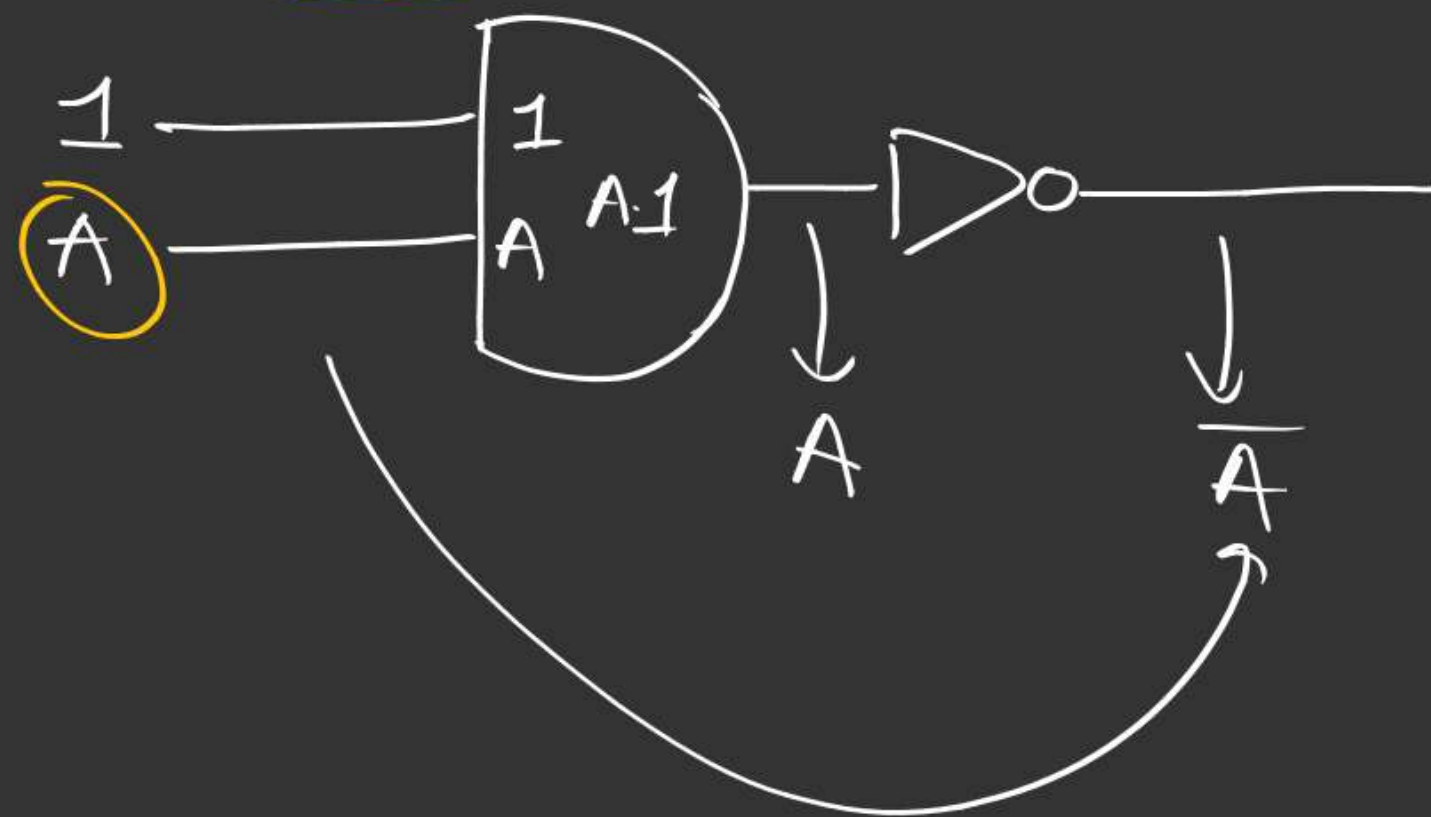
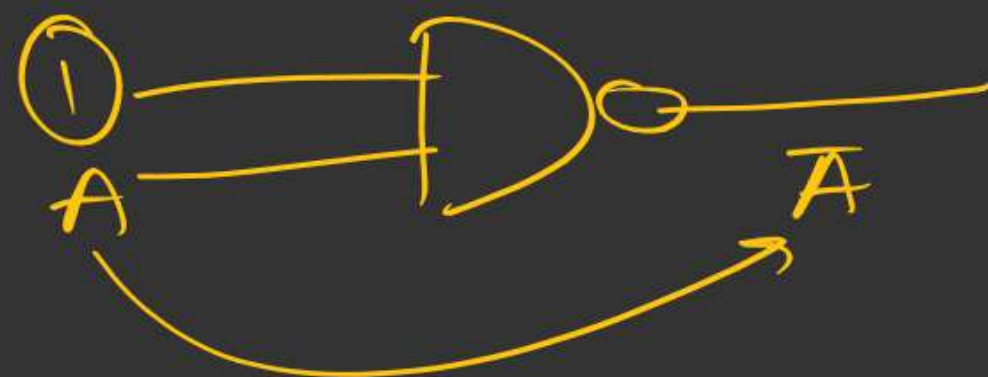
NAND



X	Y	$\overline{X \cdot Y}$
0	0	1
0	1	1
1	0	1
1	1	0

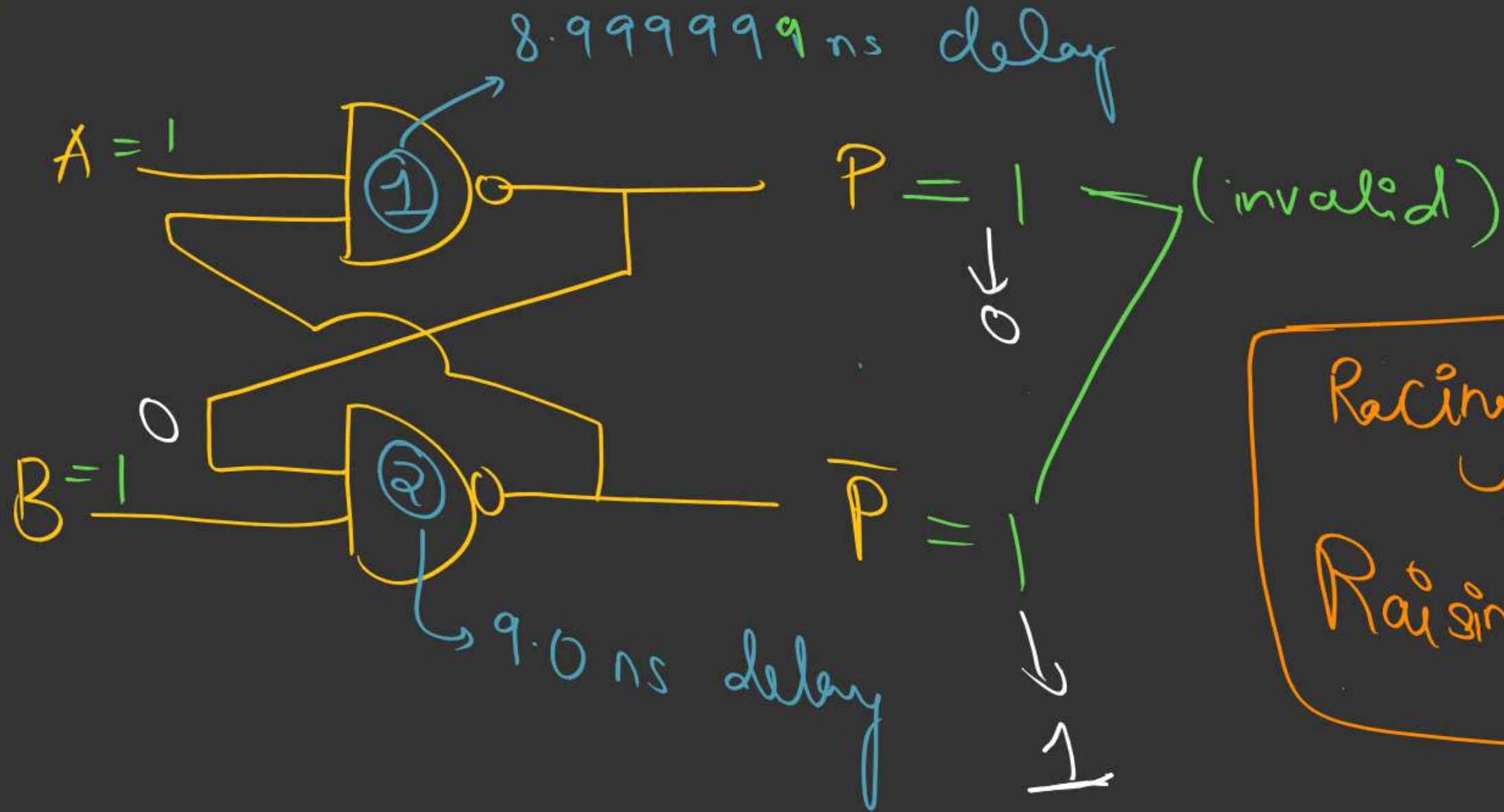
$\rightarrow$ only when all i/p's are 1, NAND gives 0.

$\rightarrow$ even if one of the i/p is 0, the o/p of NAND is 1.



No two gates has exact same delay.

Obs: [1 1 as i/p when previous o/p is invalid]



Racing Problem
OR
Raising Problem

Latches:-

Raising or Racing Problem

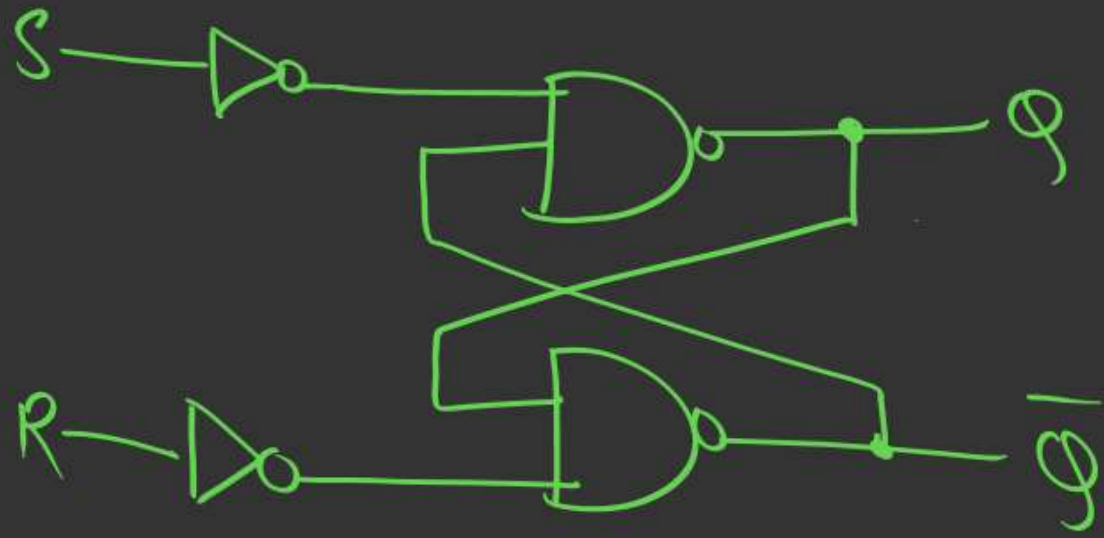
→ This problem occurs whenever both i/p's are 1
i.e. $A=1$ and $B=1$ and previous state is an invalid state.^{Condition/}

In such a scenario,

The NAND gate with lower propagation delay first changes its output whereas, the other NAND remains on its previous state.

$$\begin{bmatrix} \text{Set} \rightarrow 1 \\ \text{Reset} \rightarrow 0 \end{bmatrix}$$

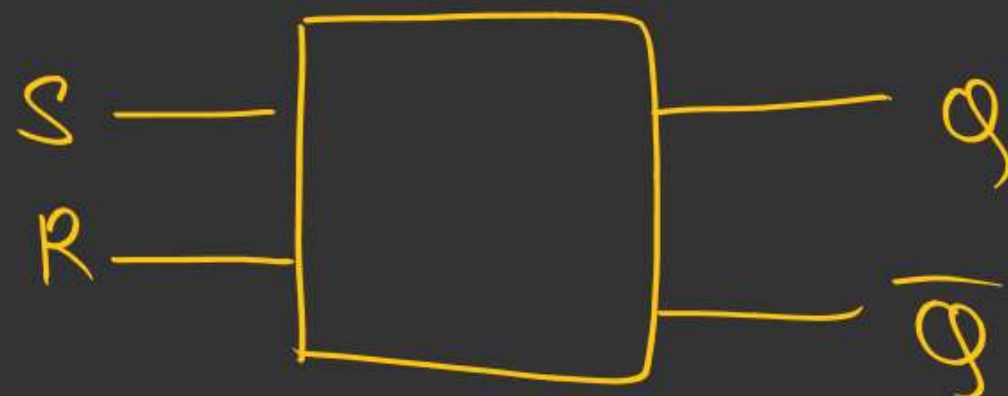
1) S-R Latch (Set-Reset Latch)



S	R	Q	$\bar{Q}$	
0	0	Q	$\bar{Q}$	→ (HOLD)
0	1	0	1	→ (Reset)
1	0	1	0	→ (Set)
1	1	X	X	→ (invalid)

Example to Remember per case

Twilight



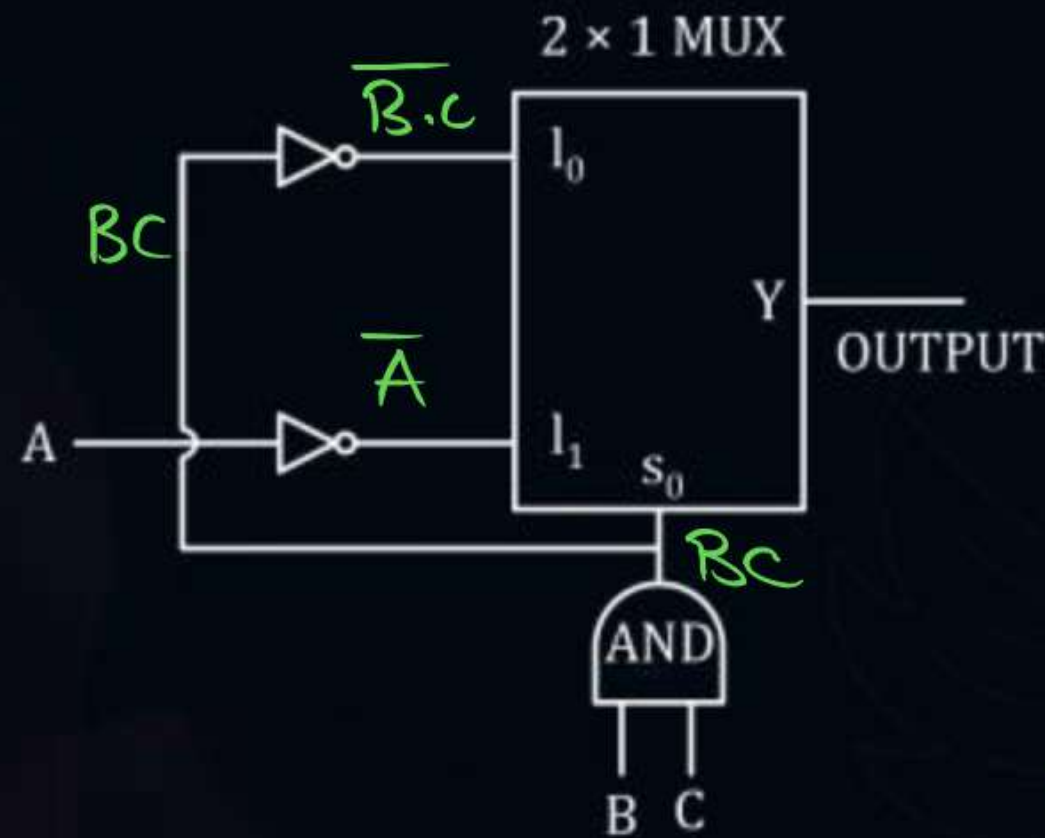
[MCQ]



#Q. The combinational circuit given below implements which of the following:

46%

- A** NOR gate
- B** XOR gate
- C** NAND gate
- D** None of these



$$S_0 = BC$$

[MCQ]



#Q. The combinational logic circuit shown in the given figure has an output Q which is-

Standard

80% → 75%

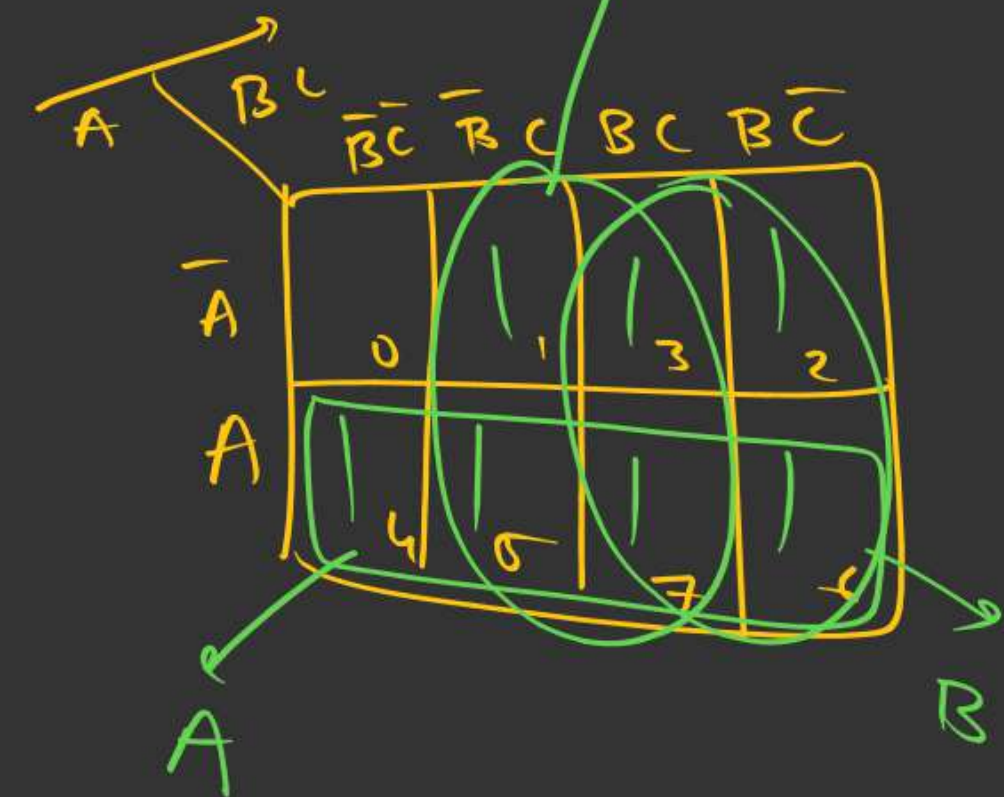
- A** ABC
- B** $A+B+C$
- C** $A \oplus B \oplus C$
- D** $A \cdot B + C$



$$\begin{aligned} Q &= \bar{A}\bar{B}C + \bar{A}B \cdot 1 + A\bar{B} \cdot 1 + AB \cdot 1 \\ &= \bar{A}\bar{B}C + B(\bar{A} + A) + A\bar{B} \\ &= \bar{A}\bar{B}C + B + A\bar{B} \\ &= \bar{A}\bar{B}C + B + A \end{aligned}$$

(lengthy)

Boolean using Kmap:



$$A+B+C$$

$$\overline{A}\overline{B}C + A\overline{B} + \overline{A}B + AB$$

001	101	010	110
↓	100	011	111
1	5,4	3,2	6,7

$$\Sigma m(1, 2, 3, 4, 5, 6, 7)$$

[MCQ]

#Q. A logic function 'f' is implemented by the circuit shown in the figure below. The circuit consists of one 2×4 decoder, two 2×1 multiplexers and a two input or gate connected in cascade. Then the function if is equal to.

H.W

A

$$A \oplus B$$

B

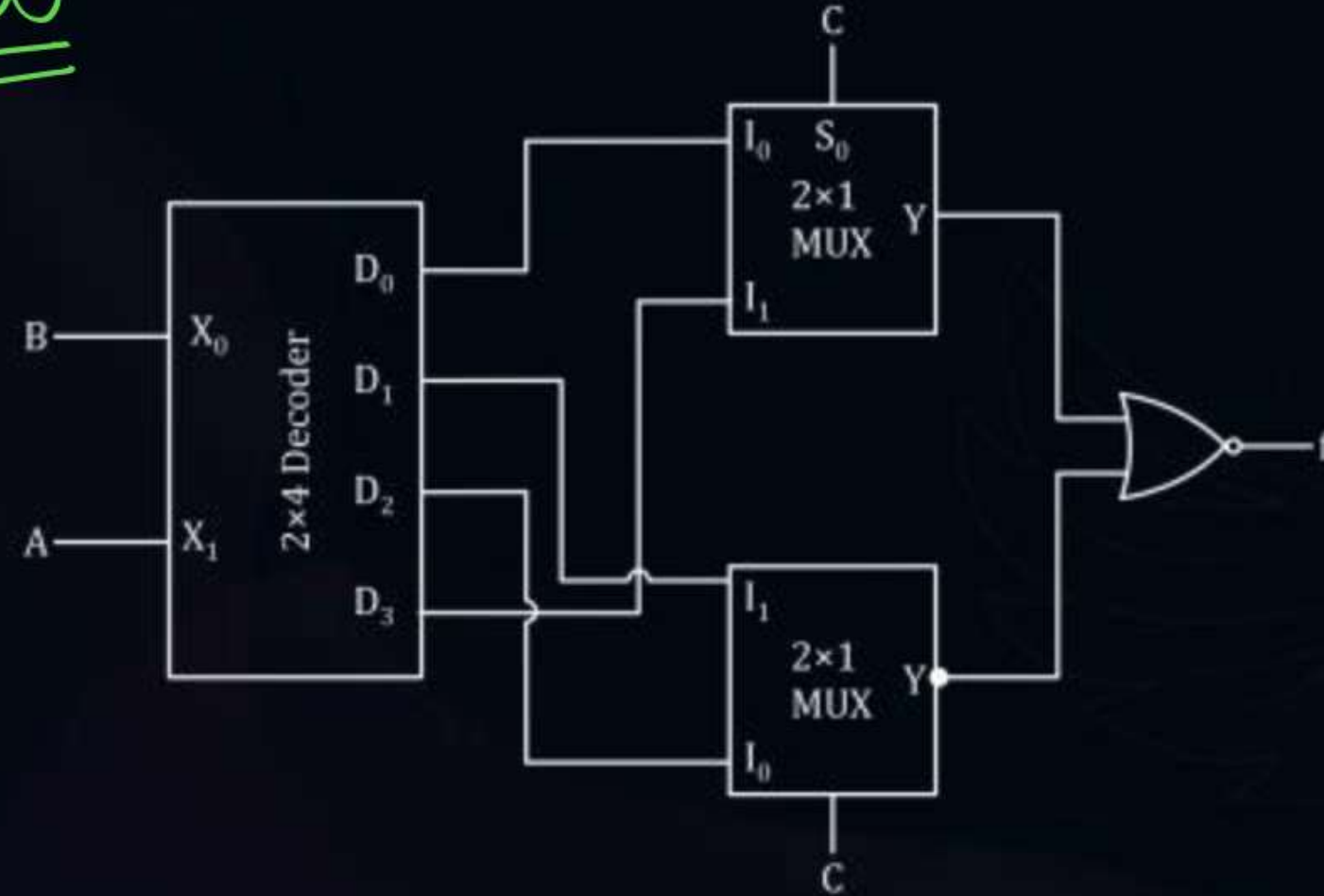
$$A \oplus B \oplus C$$

C

$$B \odot C$$

D

$$A \odot B$$





2 mins Summary



- Recap
- Sequential ch1
- Imp Questions.



THANK - YOU